

Engineering Handbook – Volume 07

Deployment & DevOps Handbook (Version 1.0)

Purpose

Defines deployment architecture, DevOps workflow and operational practices for the Energy Management SaaS platform.

Infrastructure

Current deployment uses Ubuntu on DigitalOcean with Docker, Portainer, PostgreSQL and FastAPI containers.

Container Strategy

Every major service should run in an isolated container. Docker Compose orchestrates development and production deployments.

Environment Management

Use environment variables for secrets, database credentials, JWT configuration and service endpoints. Never commit secrets to Git.

Git Workflow

Development occurs locally, changes are committed to Git, pushed to GitHub and deployed to the server using git pull and container restart.

Backup Strategy

Automate PostgreSQL dumps, retain configuration backups and version documentation. Verify restore procedures regularly.

Monitoring

Track container health, API availability, database utilization, storage, CPU, RAM and edge gateway connectivity.

Logging

Centralize application logs, API logs, MQTT logs and deployment events. Maintain audit logs for operational troubleshooting.

CI/CD Roadmap

Future pipeline: GitHub Actions → Automated Tests → Docker Build → Image Registry → Staging Deployment → Production Approval.

Disaster Recovery

Document infrastructure, automate backups, maintain rollback images and keep configuration under version control.

Operational Checklist

Verify backups, confirm database health, inspect container status, review logs, validate API endpoints, check gateway communication and monitor system resources after every deployment.